Kyle Schwarber — The State of the Swing

Use case `uc-pos-009-schwarber-swing-decay-001` · **Build** `dp_uc32` · **Ledger** UC #33
Prepared for Kellen Short (Data Product Owner) · **Value stream** `pos` — Phillies position players
Evidence window 2026-03-26 → **2026-08-07** (T-1) · career backfill 2015 → 2021 via `schwarber.parquet`
Entity lock `batter == 656941` · regular season only · 24,891 pitches · **Certification** 24/24 DQ, 59/59 verification

> **Read this first.** Every number below was computed by `dp_uc32_schwarber_swing_decay.py` in this session and traces to a CSV receipt in `out/`. Bat-tracking metrics exist only from 2024 (bat speed) and 2025 (swing path). Seasons before those dates are printed as *not measured* — never as an estimate. See §7.

Bottom line

The bat is fine. The decisions are not.

Kyle Schwarber's bat speed in 2026 is **74.2 mph** — identical to 2025 (74.2) and within a rounding error of 2024 (75.0). His 90th-percentile swing is **81.0 mph** in both halves of this season. His swing *shape* — attack angle, path tilt, swing length — is statistically indistinguishable from last year. There is no evidence in the bat-tracking data of physical decline.

What has changed is **what he swings at, and where the ball goes when he hits it.**

- **Chase rate is 25.5%** — his highest in a full season since 2017, up from 21.5% last year.
- **Strikeout rate is 34.8%** — a career high, up from 27.4%.
- Since **May 27**, his barrel rate has fallen from **24.2% to 9.8%** — a 59.5% collapse — while his bat speed moved by **0.006 mph**.

The mechanism is a **launch-angle redistribution, not a power loss**. He is still hitting the ball hard; he is hitting it hard at the wrong angles. Contact in his home-run band (20–32°) fell from 21.7% of balls in play to 14.9%. Contact in the line-drive band (8–20°) rose from 18.3% to 27.3%. He traded home runs for singles and doubles.

And a warning about the metric you asked for. Sweet-spot % — the standard 8–32° band — went **up** during the collapse, from 40.8% to 43.4%. So did hard-hit rate (50.8% → 53.3%) and squared-up rate (56.0% → 62.5%). Every "is he making good contact" metric improved while his slugging fell 27%. **Sweet-spot % is the wrong instrument for this hitter** and the report explains why in §3.

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1. Is the season actually bad? Mostly no — and that matters

Before diagnosing a decline, price the baseline honestly.

Season	PA	HR	SLG	ISO	OPS	xwOBAcon	Barrel %	Hard-hit %	EV90	Bat speed
2022	666	46	.504	.286	.824	.501	20.1%	54.4%	109.6	*not measured*
2023	716	47	.473	.277	.811	.459	16.4%	48.8%	108.1	*not measured*
2024	689	38	.484	.237	.845	.489	15.6%	55.5%	109.2	75.0
2025	719	56	**.561**	**.322**	**.918**	**.536**	**20.8%**	**59.6%**	**109.8**	74.2
2026	494	33	.518	.276	.878	.519	16.9%	52.1%	108.9	**74.2**

Receipt: `a1_career_season_spine`

2026 is a normal Schwarber season. SLG .518 and ISO .276 sit above his 2023 and 2024 marks. 33 home runs in 494 PA is a 46-homer pace over a full year. The comparison that makes 2026 look alarming is **2025 — the best offensive season of his career.** Some of the gap you are feeling is regression from a career year, not decay.

Two numbers are genuinely bad, and neither is about power:

- **Strikeout rate .348** — a career high. Prior high in a meaningful sample was .309 (2017).
- **Chase rate 25.5%** — up 4.0 points year over year, his worst full season since 2017.

Hold those. They are the thread.

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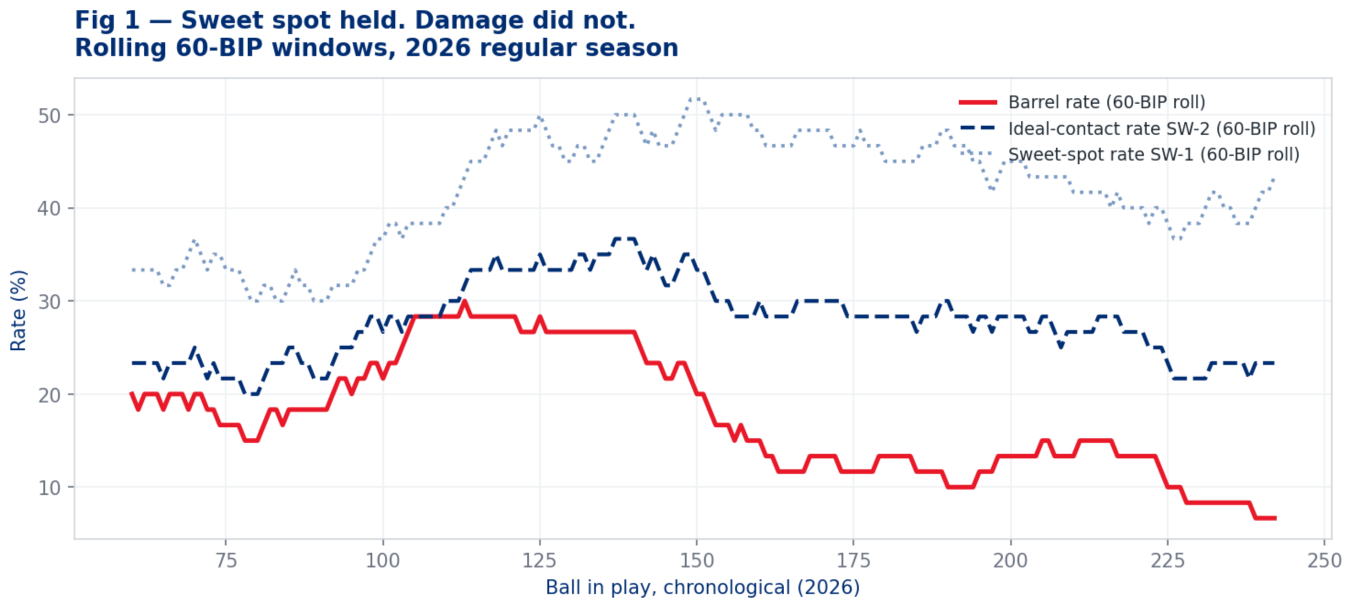
2. Within the season, the decline is real and it is steep

You said the pop has drained *as the season has progressed*. That is correct, and it is sharper than the season line suggests.

The build splits 2026 at the chronological midpoint of balls in play — **2026-05-27** — so both phases carry equal contact weight (120 BIP vs 122 BIP).

Metric	Through May 27	Since May 27	Change
Plate appearances	235	259	—
Balls in play	120	122	—
Barrel rate	**24.2%**	**9.8%**	**–59.5%**
ISO	**.372**	**.187**	**–49.7%**
SLG	.603	.439	–27.2%
OPS	.952	.810	–14.9%
xwOBA on contact	.551	.487	–11.6%
Ideal-contact rate (SW-2)	28.3%	25.4%	–10.2%
Mean launch angle	23.9°	19.1°	–4.8°
—	—	—	—
Bat speed	**74.249**	**74.243**	**–0.006 mph**
Bat speed, 90th pct	81.07	81.00	–0.07 mph
Fast-swing rate (≥75 mph)	68.1%	66.7%	–2.1%
Swing length	7.438 ft	7.441 ft	+0.0%
Attack angle	14.5°	15.2°	+4.7%
Swing path tilt	29.2°	29.5°	+1.1%
—	—	—	—
Sweet-spot rate (SW-1)	40.8%	43.4%	**+6.4%**
Hard-hit rate	50.8%	53.3%	**+4.9%**
Squared-up rate (SW-4)	56.0%	62.5%	**+11.6%**
—	—	—	—
Chase rate	24.1%	26.7%	+10.8%
Whiff rate	32.9%	36.6%	+11.5%
In-zone whiff rate	22.7%	24.4%	+7.5%

Receipts: `b5_phase_split_2026` , `b6_phase_delta`



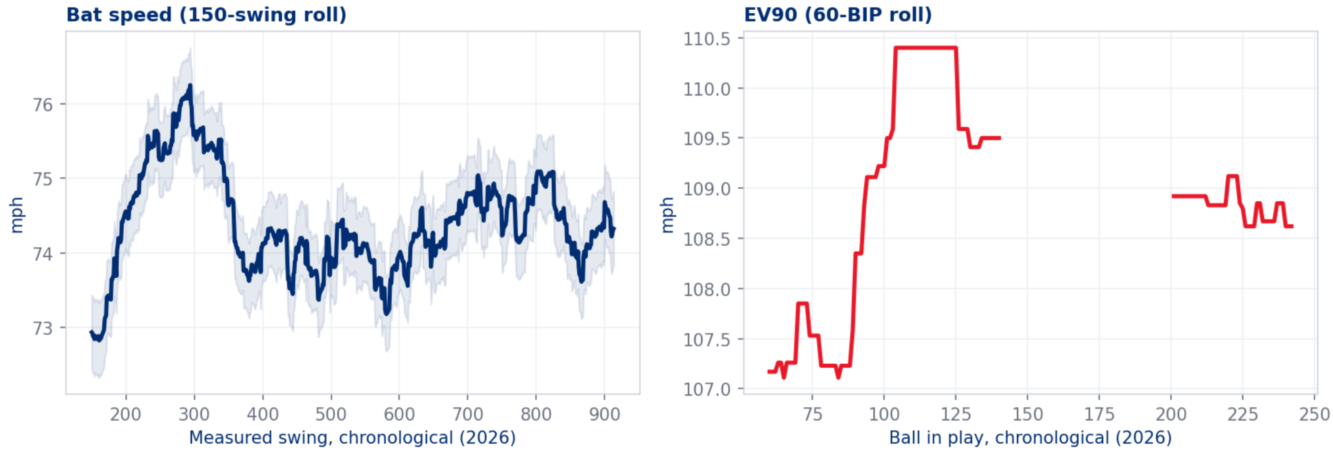
Read the middle block again. Bat speed moved six thousandths of a mile per hour. The swing did not get slower, shorter, or flatter. Whatever is happening, it is not in the engine.

Monthly, the shape is unambiguous:

Month	PA	BIP	Barrel %	SLG	HR	Bat speed	Chase %
Mar	22	10	30.0%	.474	2	71.8	24.1%
Apr	118	61	19.7%	.613	9	74.8	20.4%
May	112	57	26.3%	.592	11	74.1	30.3%
Jun	115	58	12.1%	.558	8	74.1	25.2%
Jul	95	45	8.9%	.385	3	74.4	25.3%
Aug	32	11	0.0%	.160	0	74.6	29.9%

Receipt: `b1_monthly_2026` . **August is 11 balls in play — directional only, do not read the .160.**

Fig 2 — The engine is intact; the transfer is not.



3. The mechanism: he is hitting it hard at the wrong angle

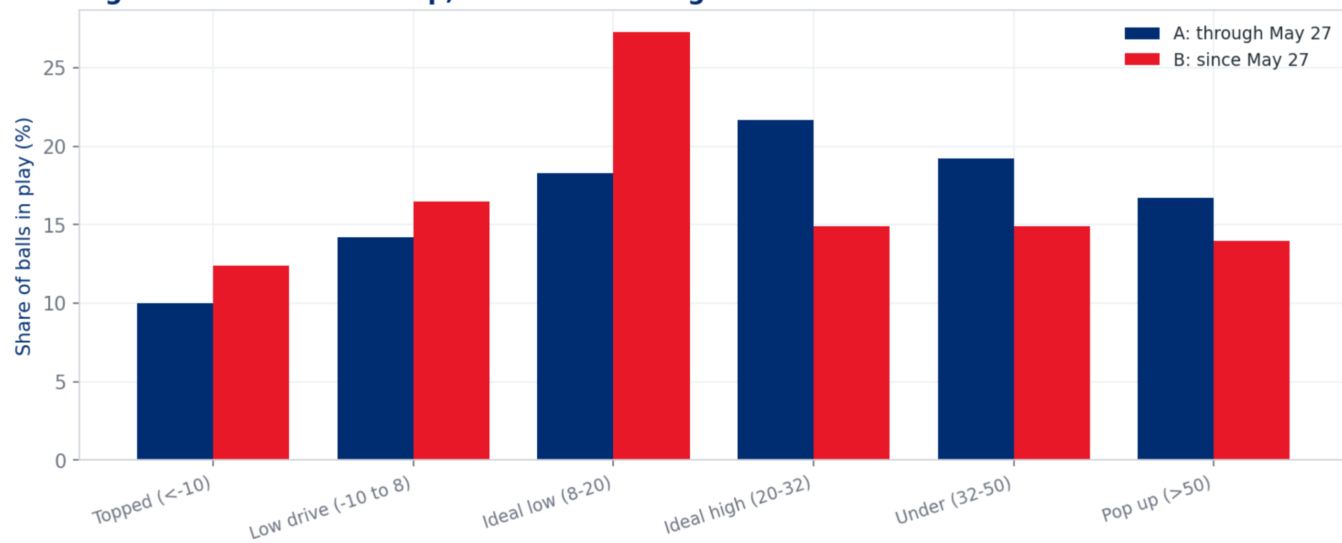
This is the finding.

Balls in play, bucketed by launch angle, with the xwOBA each bucket actually produced:

Launch-angle band	xwOBAcon	Share, through May 27	Share, since May 27	Change
Topped (< –10°)	.088	10.0%	12.4%	+2.4 pts
Low drive (–10 to 8°)	.266	14.2%	16.5%	+2.3 pts
Ideal low (8–20°)	.736	18.3%	**27.3%**	**+9.0 pts**
Ideal high (20–32°)	**1.243**	**21.7%**	**14.9%**	**–6.8 pts**
Under (32–50°)	.515	19.2%	14.9%	–4.3 pts
Pop up (> 50°)	.008	16.7%	14.0%	–2.7 pts

Receipt: `c1_la_distribution` *. xwOBAcon shown from the Phase A window.*

Fig 3 — Contact moved up, out of the damage band.



The **20–32° band is where Kyle Schwarber's value lives.** It produced an xwOBA on contact of **1.243** — it is, essentially, his home-run band. He lost nearly a third of it (21.7% → 14.9%). What he gained was the 8–20° band at **.736** — good contact, worth roughly 60% as much.

This is why sweet-spot % misled. The standard band is 8–32°. An 8° line drive and a 30° fly ball both count as "sweet spot." For a league-average hitter that is a reasonable simplification. For a hitter whose entire offensive identity is the top third of that band, it hides exactly the movement that matters. His sweet-spot % **improved** while his slugging fell 27%.

Recommendation: for this hitter, replace sweet-spot % in any dashboard or report with the **20–32° share** — call it *Damage-Band Rate* — or with **SW-2 Ideal-Contact Rate** (8–32° *and* EV ≥ 95), which at least requires the contact to be hard. SW-2 caught the decline (–10.2%); SW-1 did not. This is logged as **OI-2** for glossary ratification.

Where it shows up

By pitch group — the loss is concentrated on breaking balls:

Pitch group	Phase	PA	BIP	Barrel %	xwOBAcon	Whiff %
Breaking	through May 27	81	35	25.7%	.602	41.5%
Breaking	since May 27	99	41	**4.9%**	**.378**	**47.2%**
Fastballs	through May 27	115	67	22.4%	.520	23.0%
Fastballs	since May 27	130	73	11.0%	.528	27.2%
Offspeed	through May 27	38	18	27.8%	.566	41.8%
Offspeed	since May 27	27	8	25.0%	.684	37.0%

Receipt: `c2_pitch_group_phase`*. Offspeed Phase B is 8 balls in play — directional only.*

Breaking-ball plate appearances rose from 81 to 99 while his barrel rate against them fell to 4.9% and his whiff rate climbed to 47.2%. **Opposing clubs found something and are leaning on it.**

By velocity — it is not a velocity problem. Barrel rate fell across every band:

Velocity band	Phase A barrel %	Phase B barrel %	Bat speed A	Bat speed B
< 88 mph	27.7%	13.2%	72.2	72.1
88–93	21.9%	8.9%	75.7	75.7
93–96	21.4%	4.5%	76.8	76.3
96+	23.1%	11.8%	74.2	74.9

Receipt: `c4_velocity_band_phase`

He is still swinging **hardest at the hardest pitches** (76.3 mph vs 93–96). He is not getting beaten by velocity. The damage loss is uniform across speeds — consistent with a timing/angle issue, not a reaction-time issue.

By direction — the pull-side power went first:

Direction	Phase	BIP	Share	Barrel %	xwOBAcon
Pull	through May 27	75	63.0%	24.0%	.555
Pull	since May 27	71	58.2%	**12.7%**	.529
Straightaway	through May 27	19	16.0%	31.6%	.798
Straightaway	since May 27	33	27.0%	9.1%	.514
Oppo	through May 27	25	21.0%	16.0%	.292
Oppo	since May 27	18	14.8%	**0.0%**	.274

Receipt: `c6_spray_direction_phase`

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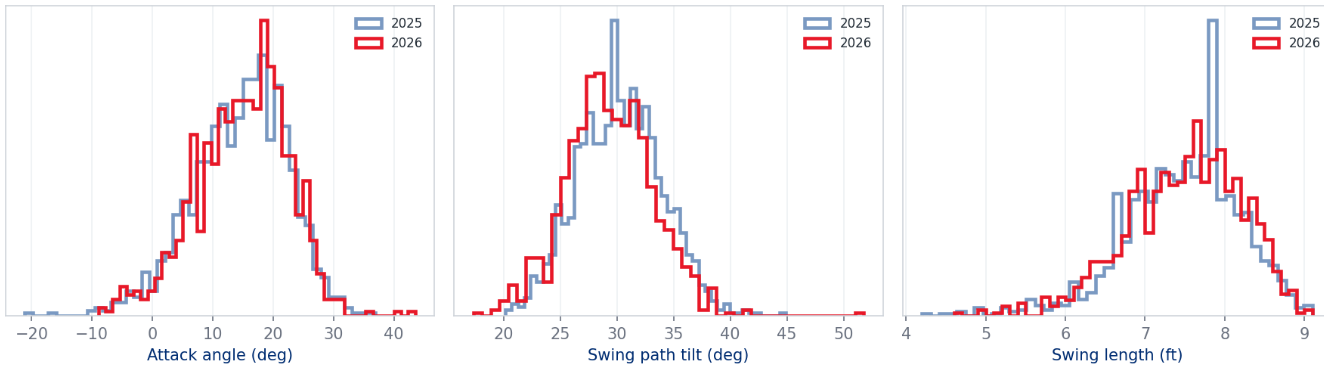
4. Swing path — what the new tracking data says

2025 is the first season with attack angle, attack direction and swing-path tilt. That gives exactly **one prior season** of comparison. Treat it as a year-over-year check, not a trend.

Metric	2025	2026	Coverage 2026
Attack angle	14.5°	14.9°	98.1% of swings
Swing path tilt	30.2°	29.3°	98.1%
Swing length	7.46 ft	7.44 ft	98.1%
Attack-angle fit rate (5–20°)	—	62.7% (Phase B)	98.1%
Contact depth	31.98 in	32.90 in	97.5% of BIP

Receipts: `d1_swing_path_year` , `d4_contact_depth` , `a2_bat_tracking_coverage`

Fig 5 — Swing shape is unchanged year over year.



The swing is the same swing. Attack angle up 0.4°, tilt down 0.9°, length unchanged. The distributions overlap almost completely.

Contact depth deserves an honest caveat. Phase A contact averaged **33.79 inches** out front; Phase B **32.03 inches**. That reads as "he lost 1.8 inches of extension." But his **2025 full-season average was 31.98 inches** — meaning **Phase B is his normal, and Phase A was the anomaly.** He was hitting the ball unusually far out front through May, which is precisely the mechanical condition that lifts launch angle into the 20–32° band. He did not break; he stopped doing something exceptional.

This is the most important interpretive point in the report, and it is why "Phase A → Phase B" should not be read as a straight decline. **Phase A was a heater above even his career-best 2025** (24.2% barrel rate vs 20.8% for 2025 as a whole). Phase B (9.8%) is genuinely below his career norm. The truth is in between, and the season line (16.9%) reflects it.

Attack angle actually predicts his damage

Attack angle	2026 BIP	Mean LA	Barrel %	xwOBAcon
< 5°	18	5.7°	0.0%	.467
5–10°	47	22.0°	17.0%	.455
10–15°	81	19.0°	18.5%	.545
15–20°	74	26.8°	18.9%	.554
20–25°	13	20.4°	7.7%	.391

Receipt: `d3_attack_angle_outcome`

His productive window is **10–20° of attack angle**, and he is in it 62.7% of the time in Phase B, down from 66.9% in Phase A. Small, but directionally consistent with everything else.

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5. Peer context — Phillies left-handed batters

Secondary framing, included because your original query set it up. Pool: Phillies LHB, Statcast era, ≥100 PA in the season.

Among the **five** Phillies LHB with measured bat tracking in 2026:

Metric	Schwarber	Pool median	Percentile
Bat speed	74.2	69.9	80th (highest)
Fast-swing rate	67.4%	15.4%	80th (highest)
Barrel rate	16.9%	8.6%	80th (highest)
EV90	108.9	103.6	80th (highest)
Ideal-contact rate	26.9%	20.6%	80th (highest)
Sweet-spot rate	42.1%	38.4%	60th
Squared-up rate	**59.3%**	**65.3%**	**0th (lowest)**
Swing path tilt	29.3°	31.0°	20th

*Receipt: `e2_lhb_percentiles_2026` . Pool n = 5. Percentiles from a five-player pool are labels, not statistics.***

He is still the hardest and fastest swinger on the roster by a distance. The one place he ranks last is **squared-up rate** — the share of contact that converts the available bat-and-pitch energy into exit velocity. That is the efficiency metric, and it is consistent with the launch-angle story: he is generating the energy and not fully converting it.

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6. What this is, and what it is not

It is not a bat-speed decline. Three independent measures (mean, 90th percentile, fast-swing rate) are flat across the season and flat year over year.

It is not a swing-mechanics change. Attack angle, path tilt and swing length are unchanged within measurement noise.

It is not a velocity problem. He swings hardest at premium velocity and his damage fell uniformly across all speeds.

It is a **swing-decision** problem compounded by a **launch-angle** drift:

1. Chase rate up 4.0 points year over year to a nine-year high. Chasing pulls contact deeper and flatter.
2. Strikeout rate at a career-high 34.8%, so fewer balls in play to do damage with.
3. Breaking-ball damage collapsed (25.7% → 4.9% barrel) while breaking-ball usage against him rose — opponents have adjusted and he has not adjusted back.
4. Contact drifted out of the 20–32° band that carries his value, back toward his 2025 baseline depth.

Sample-size honesty: Phase B is 122 balls in play. A 122-BIP barrel-rate estimate has wide error bars. The *direction* is corroborated by four independent measures (barrel, ISO, xwOBAcon, damage-band share) and by the monthly series, which is why the report treats it as real. The *magnitude* should be expected to regress.

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7. On the NULL question you raised

You imputed Schwarber's mean bat speed into the missing values and were unsure. **The instinct to question it was right, and imputation here would have been actively harmful.** The DPO decision recorded for this build is **no imputation, coverage gate**.

Measured coverage, by season:

Season window	Bat speed	Swing path	Status
2015–2023	0.0%	0.0%	**not measured**
2024	93.1%	0.0%	bat tracking only
2025	99.1%	99.1%	bat tracking + swing path
2026	98.1%	98.1%	bat tracking + swing path

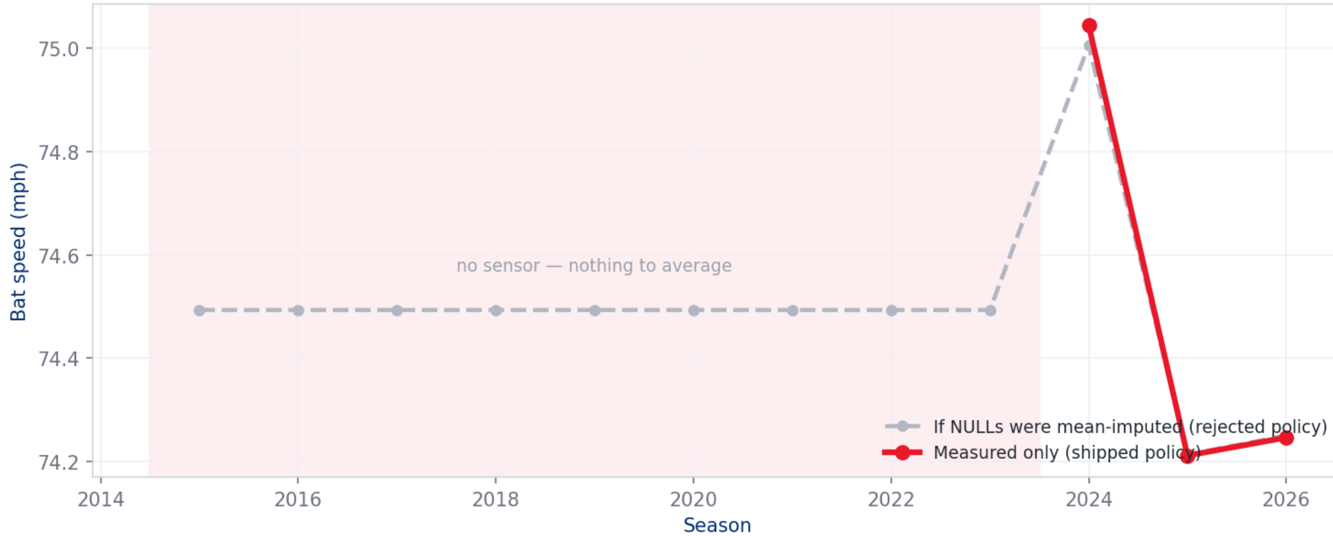
Receipt: `a2_bat_tracking_coverage`. Note: 2023 coverage is 0.0%, not "limited." Your note suggested partial 2023 availability; for this batter in this dataset there is none.

What mean-imputation would have done, quantified:

	Value
Career mean that would have been imputed	74.49 mph
Seasons with zero coverage	**9**
Swings that would have received a fabricated value	**7,021**
Share of career swings fabricated	**67.7%**
Measured standard deviation, 2026	10.06 mph

Receipt: `x1_imputation_harm`

Fig 4 — Imputation would have drawn a nine-season trend that was never measured.



Two-thirds of the career series would have been invented, and every fabricated season would have landed at exactly 74.49 mph with zero variance. The chart would have shown a perfectly flat bat-speed line from 2015 to 2023, then real movement from 2024 — and a reader would reasonably conclude "his bat speed has been rock-steady for a decade." That conclusion would be an artifact of the fill, not a finding.

The general rule this establishes for the repository (proposed as a governance standard, logged as **OI-1**):

> When a field is missing because the **instrument did not exist**, that is not missing data — it is **out-of-scope data**. Imputation is only defensible when a value existed and was not captured. Sensor-era fields (`bat_speed` , `swing_length` from 2024; `attack_angle` , `attack_direction` , `swing_path_tilt` , `intercept_*` from 2025) must be computed on measured rows only, must publish coverage alongside every aggregate, and must render pre-sensor periods as *not measured* rather than as a number.

Mechanically enforced in this build by four DQ rules — **DQ-08, DQ-09, DQ-10, DQ-11** — which fail the build if any pre-sensor value is published. All four pass.

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8. Personas and the actions available to them

Seven personas in the value stream. Each gets the finding that is actionable *for them*, and the lever they actually control.

8.1 Hitting Coach / Hitting Strategist

The finding: Not a mechanics problem. Do not rebuild the swing.
Levers:
- **Do not chase attack angle.** 14.9° is where he was in his career-best season. Any cue that steepens or flattens the path is solving a problem that does not exist.
- **Work contact depth, not swing shape.** The 20–32° band returns when he meets the ball further out front. Phase A shows what 33.8 inches produces. Cues: earlier lower-half initiation, timing work against breaking-ball speeds — not swing-plane drills.

- **Breaking-ball plan is the priority.** 4.9% barrel, 47.2% whiff against breaking balls in Phase B. Rebuild the recognition and take strategy there before touching anything else.
KPI to hold him to: Damage-Band Rate (20–32° share), not sweet-spot %. Target: back above 20%.

8.2 The Player

The finding: Your bat is exactly as fast as it was in your best season. Nothing has been lost physically.
Levers:
- The problem is **which pitches**, not **how hard**. Chase rate is the single number to move: 25.5% back toward 21.5% recovers most of the damage without touching the swing.
- Fewer two-strike counts. Two-strike PAs carry a .199 SLG in Phase B; every chase earlier in the count feeds that.
KPI: chase rate, weekly.

8.3 Advance Scouting / Game Planning (own club)

The finding: Opponents have a plan and it is working. Breaking-ball usage up, damage down 80%.
Levers:
- Reverse-engineer the specific breaking-ball shapes and locations doing the damage; build the counter-plan into pre-series prep.
- Feed the pattern to the hitting group as *scouting intelligence*, not as a mechanical fault.
KPI: breaking-ball barrel rate and whiff rate, by series.

8.4 Manager / Lineup Construction

The finding: Season-level production (.518 SLG, .878 OPS, 46-HR pace) remains top-of-lineup quality. The slump is real but the floor is high.
Levers:
- Resist a reactive lineup demotion on 122 balls in play. The corroborating evidence points to regression, not cliff.
- **Do** consider matchup protection against heavy breaking-ball staffs while the plan is rebuilt.
- Rest days are a legitimate lever — see 8.6 before using them.
KPI: rolling 60-BIP barrel rate (Fig 1) as the trigger, not the box score.

8.5 Front Office / Roster & Contract

The finding: The single most valuable input to a valuation decision — **bat speed** — shows zero decline. This is not an aging curve.
Levers:
- Any projection model that reads a 2026 power decline as physical decay is mis-specified for this player. Feed it the bat-tracking evidence.
- The risk is **contact quality and strikeout rate**, which are volatile and often recoverable, not **bat speed**, which is not.
Caution: the 34.8% strikeout rate is a career high and *is* a legitimate ageing-adjacent signal. Do not dismiss it because the bat speed is intact.
KPI: K rate and chase rate over the next 200 PA.

8.6 Performance / Sports Science

The finding: No fatigue signature in the bat-tracking data. Bat speed is flat month over month (74.8 / 74.1 / 74.1 / 74.4 / 74.6 from April to August) — the opposite of what accumulated fatigue looks like.
Levers:

- Do not attribute this to workload without independent evidence; the swing-speed data actively argues against it.
- If rest is used, it should be justified on decision-making/recovery grounds, not power output.

KPI: monthly bat speed and fast-swing rate — currently a clean bill of health, worth continuing to monitor as the falsification test.

8.7 Opposing Advance Scout (mirror view — what they already know)

Included because knowing what the other side sees is itself actionable.
- Chase rate up 4 points. **Expand the zone earlier.**
- Breaking balls: 47.2% whiff, 4.9% barrel. **Increase usage.**
- Oppo-field barrel rate is 0.0% in Phase B. **He cannot punish the outer third right now.**

The counter-plan writes itself from this list. Anything the hitting group does should start by neutralising these three.

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9. What would change this read

Three falsifiable projections. Re-run at **150 additional plate appearances** (roughly 2026-09-10):

1. **Bat speed stays at 74 ± 0.5 mph.** If it drops below 73, this report's central claim is wrong and the aging interpretation returns.
2. **Damage-Band Rate (20–32°) recovers above 18%.** If it stays below 15% with bat speed intact, the problem is durable mechanical timing, not a slump.
3. **Chase rate falls below 24%.** If it does not, the decision-making explanation hardens and the intervention should shift entirely to approach.

If (1) fails, supersede this UC rather than amend it.

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10. Candid caveats

- **Phase B is 122 balls in play.** Every rate in the phase split carries a wide interval. Direction is corroborated; magnitude will regress.
- **Phase A was a heater**, not a baseline. Its 24.2% barrel rate exceeded his career-best full season. Reading A → B as a pure decline overstates the fall by roughly half.
- **Swing path has exactly one comparison season.** 2025 vs 2026 is a year-over-year check. It cannot establish a trend.
- **The LHB peer pool has five players with measured bat tracking.** Percentiles from n=5 are descriptive labels.
- **August is 11 balls in play.** It appears in the monthly table for completeness and carries no weight.
- **Squared-up rate (SW-4) is provisional.** It applies Statcast's published max-EV formula to a plate-crossing speed derived from the 9-parameter trajectory fit. The derivation is exact physics (validated: release-minus-plate gap 7.18 mph) but the 1.23 / 0.2306 constants are published approximations. Treated as directional; see OI-3.
- **No park adjustment.** All contact-quality figures are raw.
- **No opponent-quality adjustment.** The breaking-ball finding could partly reflect who he faced.

Build: `dp_uc32_schwarber_swing_decay.py` *· 24 CSV receipts · 5 figures · DQ 24/24 · verification 59/59*
Governance trail: `00_DPO_delivery_spine.md` *through* `07_certification_and_publish_readiness.md`